A Finite Relational Kernel for Locality-Stable Extensions

Refinement, Boundary Determinacy, and Finite Certification

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Abstract

We develop a finite relational kernel built from incidence sites, finite relation schemas, and typed partial composition, with no primitive geometric, probabilistic, or continuum structure. Within this setting we define admissible extension processes, behavioral congruence, and a refinement framework governing observable compression and stabilization. The kernel is entirely finitary and supports a sharp separation between three layers of claim: unconditional finite theorems, explicitly conditional locality and boundary-control results, and certified finite-instance computations.

Our first contribution is a self-contained finite structural theory of compression under admissible tests. This yields well-defined observational quotients, singular projection interfaces, and defect-ledger bookkeeping without appeal to continuum or analytic input. Our second contribution is a locality-stable extension package, formulated through an explicit LS-2 hypothesis, under which observable outcomes in a region are determined by stabilized boundary-collar data. From this we derive boundary-controlled counting bounds for admissible local outcomes. Our third contribution is a finite certification framework: fixed-parameter computational audits are recorded as certificates and used only in the precise regimes they verify.

Under an additional explicit Phase-A boundary-capacity condition, the boundary-counting framework implies a packing contradiction restricting allowable local configurations. These consequences are conditional and are stated only with their full dependency stack. No continuum limit, field-theoretic identification, Yang–Mills closure, or Navier–Stokes closure is claimed in the kernel theorem package. All such large-scale interpretations are isolated as separate bridge programs and are not used in the proofs of the finite results.

The result is a referee-auditable finite theorem package that separates proved structural facts, certified finite classifications, and open bridge obligations.

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1 Introduction

1.1 Aim and scope

This paper isolates the finite theorem-bearing core of the Nested Fibrational Cosmology program. Its purpose is not to present the full conceptual architecture of that program, nor to argue for any particular continuum or physical interpretation. Its purpose is narrower and more durable: to formulate and prove a finite relational kernel in which stabilization, locality-sensitive extension, boundary determinacy, and finite certification can be stated with explicit dependency control.

The starting point is austere. We assume only finite incidence sites, finite relation data, and typed partial composition. We do not assume primitive geometry, continuum structure, probability, background spacetime, or primitive identity. Observable structure is instead built from admissible finite tests and the compression they induce on the space of configurations. The resulting theory is therefore finite from the ground up: every load-bearing object appearing in the theorem package is combinatorial, finite, and explicitly licensed.

The central editorial decision of this paper is a strict separation of claim types. The first layer consists of unconditional finite structural results proved directly from the kernel primitives. The second layer consists of explicitly conditional locality and boundary results, requiring a named locality-stability package, most centrally LS-2, and in later consequences an additional Phase-A boundary-capacity condition. The third layer consists of finite computational certifications. These are not used as substitutes for general proofs. Instead, they are recorded as fixed-regime certificates verifying named obligations in explicitly stated parameter windows.

This separation is the main methodological point of the paper. The present work is designed to be read under hostile conditions. A reader should be able to determine, at every stage, whether a statement is unconditional, conditional, computational, or open.

1.2 Mathematical problem addressed here

The mathematical problem addressed in this paper is the following.

1. Can observable equivalence be induced by admissible internal tests rather than primitive identity?

2. Can stabilization be obtained through an explicit refinement mechanism?

3. Can locality be formulated so that observable data in a region are controlled by stabilized boundary information?

4. Can such control be converted into finite counting bounds and genuine structural exclusions under clearly stated capacity assumptions?

This paper gives a positive answer in a precise finite setting. The answer is not a continuum theory. It is not a solution of any field equation. It is not a derivation of the Standard Model, a proof of the Yang–Mills mass gap, or a proof of global regularity for Navier–Stokes. What is claimed here is a finite theorem package that stands independently of those ambitions.

1.3 Guiding idea

The guiding idea is that observable structure should be defined operationally, through admissible relational tests, and that the failure of canonical inversion under compression should be treated as a structural feature rather than a defect of formalization.

Once configurations are quotiented by behavioral indistinguishability, one obtains compression classes rather than token-level identities. In general, this compression is many-to-one. The absence of a canonical inverse produces singular projection interfaces. From there, the theory develops a stabilization and defect-accounting framework that tracks what survives compression, what is lost, and how locality restricts admissible extensions.

The key technical transition in the paper is from compression to boundary determinacy. Under a locality-stable extension hypothesis, local observable outcomes become determined by stabilized collar data. This converts qualitative locality into quantitative counting control. Once counting is available, additional capacity assumptions can force contradictions for over-packed local behavior. That is the theorem spine of the paper.

1.4 Main result map

The result classes of the paper can be read in the following order.

1. The finite relational kernel supports a coherent observational quotient theory.
2. Refinement chains are monotone and stabilize in finite time.
3. Under LS-2, stabilized local observables are boundary-determined.
4. Boundary determinacy yields a counting inequality controlled by collar data.
5. Under Phase-A multiplicity demands, capacity-exceeding local regimes are excluded.
6. Certificates instantiate the theorem package in named finite windows without enlarging theorem scope.

1.5 Flagship theorem block

The paper's theorem surface may be summarized as follows.

1. The finite relational kernel supports a well-defined observational quotient and singular-interface formalism.
2. Every admissible refinement chain on a finite configuration space stabilizes after finitely many steps.
3. Under LS-2, stabilized local observable outcomes are determined by bounded-radius collar data.
4. Under a finite collar alphabet, stabilized local outcome multiplicity is bounded by boundary capacity.
5. Under an explicit Phase-A multiplicity demand, capacity-exceeding local regimes are excluded.

1.6 What is not proved

Equally important is what this paper does not prove. It does not prove any continuum-limit theorem. It does not identify the kernel with a physical spacetime theory. It does not establish a quantum field theory. It does not prove the Yang–Mills existence and mass gap problem. It does not prove the Navier–Stokes existence and smoothness problem. It does not claim that certified finite computations imply asymptotic analytic closure. All such directions belong to separate bridge programs.

1.7 Status convention

Every load-bearing claim in the flagship paper belongs to exactly one of four status classes:

1. **Unconditional** — derivable from the finite kernel alone;
2. **Conditional** — dependent on explicitly named hypotheses such as LS-2 or Phase-A;
3. **Computational** — certified only in a named finite regime;
4. **Open** — not proved here and not used as though proved.

No statement is allowed to slide between these classes without explicit notice.

1.8 Organization

Section 3 defines the finite relational kernel and observational quotient. Section 4 develops refinement, survivor chains, and defect bookkeeping. Section 5 states LS-2 and proves boundary determinacy. Section 6 derives the boundary-capacity inequalities. Section 7 proves the packing contradiction under an explicit multiplicity hypothesis. Section 8 isolates the computational layer. Section 9 states the open bridge obligations.

1.9 Core notation

We reserve Ω for the finite configuration space, Σ for the observational quotient, $\mathcal{T}_n$ for the stage-n licensed test family, Σ_n for the stage-n quotient under refinement, $\partial_r U$ for the radius-r collar of a region U, ρ^* for the LS-2 collar radius, and $\mathrm{Cap}_\partial(U) = |\partial_{\rho^*} U| \log K_0$ for the boundary capacity.

2 Theorem spine

For convenience, we record the paper's theorem path in compact form.

Theorem 2.1 (Flagship theorem spine)**.** *The paper establishes the following implication chain.*

1. *Finite admissible-test quotients define a coherent observational state space.*
2. *Every finite refinement chain stabilizes.*
3. *Under LS-2, stabilized local observable outcomes are determined by stabilized collar data.*
4. *Under a finite collar alphabet, stabilized local outcome multiplicity is bounded by boundary capacity.*
5. *Under an explicit Phase-A multiplicity demand, any capacity-exceeding local regime is impossible.*

Remark 2.2. This theorem is not an additional result beyond the body of the paper. It is a roadmap theorem: each of its components is proved in the sections that follow, with the conditional steps labeled explicitly.

3 Finite Relational Kernel

This section fixes the finite kernel used throughout the paper. The aim is to make the observational quotient theory precise while refusing all hidden appeals to primitive identity, ambient geometry, or continuum structure.

3.1 Primitive stance

We work in a purely finite relational setting. No primitive spacetime, metric, probability space, or background field is assumed. No token-level identity is permitted beyond membership in a finite site set. What counts as observable must be generated internally by admissible finite processes.

Remark 3.1 (Standing convention). Throughout the paper, raw configurations are never treated as physically or observationally meaningful merely by virtue of token-level distinctness. Distinctions count only when they survive the admissible-test formalism introduced below.

3.2 Finite signatures and configurations

Definition 3.2 (Finite relational signature)**.** A finite relational signature is a finite family $\mathcal{L} = \{R_i\}_{i\in I}$ in which each R_i has finite arity $k_i \in \mathbb{N}$.

Definition 3.3 (Finite configuration)**.** A finite configuration is a finite $\mathcal{L}$-structure

$$\omega = (X, \{R_i^\omega\}_{i\in I}),$$

where X is a finite set of sites and $R_i^\omega \subseteq X^{k_i}$ for each $i \in I$.

Definition 3.4 (Configuration space)**.** A configuration space is a finite set Ω of finite configurations over the same base signature $\mathcal{L}$.

Definition 3.5 (Structural equivalence)**.** Two configurations $\omega, \omega' \in \Omega$ are structurally equivalent, written $\omega \cong \omega'$, if there exists an $\mathcal{L}$-isomorphism between them.

3.3 Extension processes and admissibility

Definition 3.6 (Extension process)**.** An extension process is a map

$$E : \Omega \to \mathbf{Str}^{\mathrm{fin}}_{\mathcal{L}_E},$$

where $\mathcal{L}_E \supseteq \mathcal{L}$ is a finite expansion of the base signature.

Definition 3.7 (Admissible class)**.** A class $\mathfrak{E}$ of extension processes is admissible if it satisfies:

(A1) closure under typed composition whenever the composition is defined,

(A2) structural invariance: $\omega \cong \omega' \Rightarrow E(\omega) \cong E(\omega')$ for all $E \in \mathfrak{E}$,

(A3) base-signature accountability: observational distinctions induced by $\mathfrak{E}$ remain readable at the base-signature level and do not smuggle external labels,

(A4) finite output control: for every $E \in \mathfrak{E}$, the image $E(\Omega)$ has finitely many isomorphism types.

Remark 3.8. The formal identity extension may serve as a composition unit, but it is not automatically licensed as an observational test. Otherwise primitive re-identification would enter at the ground floor.

3.4 Licensed internal tests and behavioral congruence

Definition 3.9 (Licensed internal test family)**.** A licensed internal test family $\mathcal{T}$ is a finite family of admissible relational processes whose outputs are evaluated only through structures already licensed at the current stage of the theory. Tests in $\mathcal{T}$ may not appeal to external naming, primitive coordinates, or hidden memory of raw labels.

Definition 3.10 (Behavioral congruence)**.** For $\omega, \omega' \in \Omega$, write $\omega \sim \omega'$ if for every test $T \in \mathcal{T}$, the outputs $T(\omega)$ and $T(\omega')$ yield the same currently licensed observable outcome, or are both undefined.

Proposition 3.11. *Behavioral congruence is an equivalence relation on Ω.*

Proof. Reflexivity and symmetry are immediate. Transitivity holds because if no licensed test separates ω from ω' and none separates ω' from ω'', then none separates ω from ω''. □

Definition 3.12 (Observational quotient)**.** The observational quotient is

$$\Sigma := \Omega/{\sim}.$$

Definition 3.13 (Projection interface)**.** A projection interface is a partial reduction map

$$\pi : \Omega \dashrightarrow \Omega_{\mathrm{red}}$$

that preserves currently licensed observable behavior while forgetting finer relational structure.

Definition 3.14 (Singular projection interface)**.** A projection interface π is singular at $y \in \Omega_{\mathrm{red}}$ if there exist distinct $x_1, x_2 \in \Omega$ with $\pi(x_1) = \pi(x_2) = y$, and no licensed inverse selection rule at y distinguishes x_1 from x_2.

Proposition 3.15. *If there exist $\omega_1 \neq \omega_2$ in Ω with $\omega_1 \sim \omega_2$, then the canonical quotient map $q : \Omega \to \Sigma$ is singular at the class $[\omega_1]_\sim = [\omega_2]_\sim$.*

Proof. Because $\omega_1 \sim \omega_2$, both map to the same quotient class. Because $\omega_1 \neq \omega_2$, the map is many-to-one there. Any admissible inverse selector would distinguish ω_1 from ω_2, contradicting behavioral congruence. □

Remark 3.16 (Kernel output). The kernel section delivers the paper's first stable objects: the observational quotient and the singular-interface formalism forced by many-to-one compression. Everything downstream is built on these objects rather than on raw token-level identity.

4 Compression, Refinement, and Stabilization

We now build the first dynamical structure on top of the observational quotient. The purpose of this section is to replace vague language about emergence or information loss with precise finite objects: refinement chains, survivor components, continuation defect, and stabilization.

4.1 Refinement chains

Let $\mathcal{T}_0 \subseteq \mathcal{T}_1 \subseteq \mathcal{T}_2 \subseteq \cdots$ be a nested sequence of licensed internal test families. Each stage induces a behavioral congruence relation $\sim_n$ and quotient space

$$\Sigma_n := \Omega/{\sim_n}.$$

Definition 4.1 (Refinement chain)**.** A refinement chain on Ω is a nested sequence of licensed internal test families $(\mathcal{T}_n)_{n\geq 0}$ such that each induced quotient Σ_n is finite.

Proposition 4.2 (Monotonicity of refinement)**.** *If $\mathcal{T}_n \subseteq \mathcal{T}_{n+1}$, then $\omega \sim_{n+1} \omega' \Rightarrow \omega \sim_n \omega'$. Equivalently, every class of Σ_{n+1} is contained in a class of Σ_n.*

Proof. If two configurations are indistinguishable by the larger test family, then they are indistinguishable by every test in the smaller family. Hence every equivalence class for $\sim_{n+1}$ lies inside a class for $\sim_n$. □

Thus refinement proceeds only by partition splitting: classes may stay the same or split into smaller classes, but they do not merge.

4.2 Finite stabilization

Definition 4.3 (Stabilization)**.** A refinement chain stabilizes at stage N if

$$\Sigma_n = \Sigma_N \qquad \text{for all } n \geq N.$$

Theorem 4.4 (Finite stabilization)**.** *Every refinement chain on a finite configuration space Ω stabilizes after finitely many steps.*

Proof. Each quotient Σ_n corresponds to a partition of the finite set Ω. By monotonicity, these partitions become successively finer. A finite set admits only finitely many partitions, so a strictly refining chain cannot continue indefinitely. Therefore there exists N such that the induced partition stops changing, i.e. $\Sigma_n = \Sigma_N$ for all $n \geq N$. □

Fix a stabilization stage N and write $\Sigma_\infty := \Sigma_N$ for the stabilized quotient. The notation Σ_∞ is mnemonic only; no infinite limit object is introduced.

4.3 Survivors and defect bookkeeping

Definition 4.5 (Survivor component)**.** Let $C_n \in \Sigma_n$. A survivor component of C_n is a chain of classes

$$C_n \supseteq C_{n+1} \supseteq C_{n+2} \supseteq \cdots,$$

with $C_m \in \Sigma_m$ for all $m \geq n$.

A survivor component records persistence through successive admissible refinements. It is not a token-level identity claim; it is a class-level persistence claim compatible with compression.

Definition 4.6 (Continuation defect)**.** Let $C_n \in \Sigma_n$. Its continuation defect at stage n is the failure of stage-n admissible data to determine a unique descendant class in Σ_{n+1}.

Definition 4.7 (One-step defect number)**.** For $C_n \in \Sigma_n$, define

$$\delta(C_n) := \#\{C_{n+1} \in \Sigma_{n+1} : C_{n+1} \subseteq C_n\} - 1.$$

Thus $\delta(C_n) = 0$ exactly when continuation is unique at the next step.

Proposition 4.8. *For every $C_n \in \Sigma_n$, one has $\delta(C_n) \geq 0$, with equality iff C_n has a unique descendant in Σ_{n+1}.*

Proof. Every element of C_n belongs to some class of the finer partition Σ_{n+1}, so at least one descendant exists. Therefore the number of descendants is at least 1, hence $\delta(C_n) \geq 0$. Equality holds exactly when there is precisely one descendant. □

Definition 4.9 (Defect ledger)**.** For a survivor chain

$$C_n \supseteq C_{n+1} \supseteq \cdots \supseteq C_m,$$

its defect ledger is

$$\Delta(C_n \rightsquigarrow C_m) := \sum_{k=n}^{m-1} \delta(C_k).$$

Proposition 4.10 (Ledger additivity)**.** *For any intermediate stage ℓ with $n \leq \ell \leq m$,*

$$\Delta(C_n \rightsquigarrow C_m) = \Delta(C_n \rightsquigarrow C_\ell) + \Delta(C_\ell \rightsquigarrow C_m).$$

Proof. Immediate from the definition of Δ as a sum of one-step defect numbers. □

Definition 4.11 (Stabilizer class)**.** A class $C_n \in \Sigma_n$ is a stabilizer class if every descendant chain issuing from C_n has zero defect from stage n onward.

Proposition 4.12. *If the refinement chain stabilizes globally at stage N, then every class in Σ_N is a stabilizer class.*

Proof. After stage N, the partitions do not change, so every class has itself as its unique descendant at every later stage. Hence every later one-step defect vanishes. □

Remark 4.13. The logical order of the paper is deliberate. First one defines the observational quotient. Next one proves finite stabilization. Only after stabilized classes exist does one ask whether boundary data determine local observable outcomes. This blocks premature locality claims.

5 Locality-Stable Extensions and Boundary Determinacy

We now introduce the first major conditional package of the paper. The central question is whether stabilized local observable outcomes can be controlled by bounded collar data. The answer is not unconditional. It is isolated under the named hypothesis LS-2.

Definition 5.1 (Region)**.** A region is a finite subset U of the site set.

Definition 5.2 (Collar of radius r)**.** A collar of radius r around U, denoted $\partial_r U$, is the finite boundary neighborhood determined by the admissible relational graph structure at distance at most r from the interface of U.

Definition 5.3 (Stabilized collar type)**.** A stabilized collar type of radius r for a region U is the stabilized observable class of the collar configuration $\partial_r U$ under the refinement program of section 4.

Hypothesis 5.4 (LS-2)**.** There exists a finite radius ρ^* such that for every admissible local question on a region U, once the refinement program has stabilized, the observable local outcome on U is determined by the stabilized collar type on $\partial_{\rho^*} U$.

Remark 5.5. LS-2 does not say that raw interior microstructure is recoverable from the boundary. It says only that the *stabilized observable outcome class* is controlled by bounded collar data. The theorem that follows is therefore an observable boundary law, not a microscopic reconstruction theorem.

Theorem 5.6 (Boundary determinacy under LS-2)**.** *Assume LS-2 with stabilization radius ρ^*. Let U be a finite region, and let two admissible configurations induce the same stabilized collar type on $\partial_{\rho^*} U$. Then they induce the same stabilized local observable outcome on U.*

Proof. Under LS-2, the stabilized observable outcome on U is determined by the stabilized collar type on $\partial_{\rho^*} U$. Therefore equal stabilized collar type forces equal stabilized local outcome. □

Corollary 5.7 (Boundary law map)**.** *Under LS-2, for every region U there exists a map*

$$\Phi_U : \{\textit{stabilized collar types on } \partial_{\rho^*} U\} \to \{\textit{stabilized local observable outcomes on } U\}$$

such that the stabilized local outcome on U is exactly Φ_U evaluated on the stabilized collar type.

Proof. If two admissible instances have the same stabilized collar type, the theorem above forces them to have the same stabilized local outcome. Hence the latter depends only on the former, defining Φ_U. □

Proposition 5.8 (Injectivity on realized outcomes)**.** *Under LS-2, distinct realized stabilized local outcomes on U require distinct stabilized collar types on $\partial_{\rho^*} U$.*

Proof. If two distinct realized outcomes arose from the same collar type, the boundary determinacy theorem would force them to be equal. □

Remark 5.9. This is the decisive transition in the paper. Once distinct local outcomes inject into collar types, counting control becomes available.

6 Boundary-Controlled Counting Bounds

Throughout this section, LS-2 is assumed. The logic is simple: distinct stabilized local outcomes require distinct stabilized collar types, while the latter are controlled by a finite collar alphabet. This yields the main quantitative bound of the paper.

Definition 6.1 (Realized outcome set)**.** Let $\mathcal{O}(U)$ denote the set of realized stabilized local observable outcome classes on U.

Definition 6.2 (Realized collar-type set)**.** Let $\mathcal{C}_{\rho^*}(U)$ denote the set of realized stabilized collar types on $\partial_{\rho^*}U$.

Theorem 6.3 (Outcome-count bound from boundary determinacy)**.** *For every finite region U,*

$$|\mathcal{O}(U)| \leq |\mathcal{C}_{\rho^*}(U)|.$$

Proof. By injectivity on realized outcomes, distinct elements of $\mathcal{O}(U)$ require distinct elements of $\mathcal{C}_{\rho^*}(U)$. Hence the cardinality bound follows. □

Definition 6.4 (Collar alphabet)**.** A collar alphabet at radius ρ^* is a finite set Σ_∂ whose symbols encode the stabilized local observational data available at individual boundary sites in $\partial_{\rho^*}U$. Its cardinality is denoted

$$K_0 := |\Sigma_\partial|.$$

Proposition 6.5. *For every finite region U,*

$$|\mathcal{C}_{\rho^*}(U)| \leq K_0^{|\partial_{\rho^*}U|}.$$

Proof. A stabilized collar type is determined by a finite assignment of alphabet symbols to the collar. The number of raw assignments is at most $K_0^{|\partial_{\rho^*}U|}$. Realized collar types form a subset or quotient of these assignments, so their cardinality cannot exceed this bound. □

Theorem 6.6 (Boundary-capacity counting bound)**.** *Assume LS-2 with stabilized collar alphabet of size K_0. Then for every finite region U,*

$$|\mathcal{O}(U)| \leq K_0^{|\partial_{\rho^*}U|}.$$

Proof. Combine the previous theorem and proposition. □

Corollary 6.7 (Interface-capacity inequality)**.** *Under the hypotheses of the preceding theorem,*

$$\log |\mathcal{O}(U)| \leq |\partial_{\rho^*}U| \log K_0.$$

Definition 6.8 (Boundary capacity)**.** The boundary capacity of U at radius ρ^* is

$$\mathrm{Cap}_\partial(U) := |\partial_{\rho^*}U| \log K_0.$$

Remark 6.9. This inequality is the main quantitative output prior to contradiction. It is a finite interface-capacity statement. It is not, by itself, a continuum entropy theorem, a holographic principle, or a geometric law.

Remark 6.10 (Dependency warning). The capacity inequality is not a kernel theorem in isolation. Its crucial injective step comes from boundary determinacy, and boundary determinacy comes from LS-2. The paper therefore treats the inequality as explicitly conditional rather than as an unconditional property of the finite kernel.

7 Phase-A Packing Consequences

We now add the explicit multiplicity hypothesis needed to convert boundary capacity into contradiction. The contradiction mechanism is completely transparent: Section 6 provides an upper bound, while Phase-A provides a lower bound. If the lower bound exceeds the upper bound, the regime is impossible.

Definition 7.1 (Phase-A multiplicity demand)**.** A region U satisfies a Phase-A multiplicity demand $M(U)$ if the admissible regime under consideration requires at least $M(U)$ distinct stabilized local observable outcome classes on U, i.e.

$$|\mathcal{O}(U)| \geq M(U).$$

Theorem 7.2 (Phase-A packing contradiction)**.** *Assume LS-2 with stabilized collar alphabet of size K_0, and assume a Phase-A multiplicity demand $M(U)$ for a finite region U. If*

$$M(U) > K_0^{|\partial_{\rho^*} U|},$$

then the corresponding admissible regime is impossible.

Proof. By the boundary-capacity counting bound, every admissible instance satisfies $|\mathcal{O}(U)| \leq K_0^{|\partial_{\rho^*} U|}$. If Phase-A requires $|\mathcal{O}(U)| \geq M(U) > K_0^{|\partial_{\rho^*} U|}$, the inequalities are incompatible. Hence no such admissible instance exists. □

Corollary 7.3 (Logarithmic contradiction criterion)**.** *If*

$$\log M(U) > |\partial_{\rho^*} U| \log K_0,$$

then the regime is excluded.

Definition 7.4 (Capacity-exceeding regime)**.** A region U is capacity-exceeding under a multiplicity demand $M(U)$ if

$$\log M(U) > \mathrm{Cap}_\partial(U).$$

Corollary 7.5 (Local exclusion theorem)**.** *Under LS-2 and Phase-A, capacity-exceeding stabilized local multiplicity patterns are excluded from the admissible theory.*

Remark 7.6. The theorem is an exclusion result, not a classification result. It rules out over-packed local observable regimes, but it does not derive a unique surviving local structure.

8 Computational Certificates and Evidentiary Boundaries

This section isolates the computational layer. A certificate may verify a named finite predicate in a named finite regime. It may not silently upgrade a finite check into a general theorem.

Definition 8.1 (Computational certificate)**.** A computational certificate is a finite machine-verifiable record asserting that a designated predicate P has been exhaustively checked and confirmed on a specified finite regime $\mathcal{R}$.

Definition 8.2 (Soundly scoped certificate)**.** A certificate is soundly scoped if its statement explicitly names the finite regime $\mathcal{R}$ and does not claim any conclusion beyond that regime.

Lemma 8.3 (Certificate instantiation lemma)**.** *Let*

$$(P_1 \wedge \cdots \wedge P_m) \Longrightarrow Q$$

be an analytically proved implication for finite regimes of a specified class. Let $\mathcal{R}_0$ be a fixed finite regime, and suppose a soundly scoped computational certificate verifies $P_1, \ldots, P_m$ on $\mathcal{R}_0$. Then Q holds on $\mathcal{R}_0$.

Proof. The certificate establishes that $P_1, \ldots, P_m$ hold on $\mathcal{R}_0$. The analytic implication then applies to $\mathcal{R}_0$. □

Remark 8.4 (Use convention)*.* A computational certificate may be used only to verify a finite predicate in a named regime, to instantiate an abstract theorem in that regime, or to support the claim that a conditional package has been checked on specific bounded examples. It may not replace an analytic proof of a universally quantified statement or justify asymptotic extrapolation without a bridge theorem.

Remark 8.5 (Repository companion)*.* The flagship paper cites certificates functionally rather than exhaustively. Full machine artifacts—including code, bounded parameter specifications, output logs, and digests—should live in a repository companion or technical certificate appendix rather than in the theorem path of the main text.

9 Bridge Programs and Open Obligations

The finite theorem package established in this paper is complete on its own terms. Any further interpretation—continuum, gauge-theoretic, Yang–Mills, Navier–Stokes, or cosmological—requires additional bridge theorems and is not claimed here.

Definition 9.1 (Bridge statement)**.** A bridge statement is any assertion connecting the finite theorem package to a mathematical or physical domain not already internal to the kernel.

Remark 9.2 (Open obligations)*.* The open frontier includes at least the following obligations:

1. **O-CONT:** a theorem connecting stabilized finite kernel data to a mathematically controlled continuum realization;
2. **O-SECTOR:** a theorem showing that finite extracted sector algebra persists in a scaling regime with stable mathematical meaning;
3. **O-YM:** a theorem stack connecting finite sector data and bounded certificates to the existence and mass-gap requirements of quantum Yang–Mills theory;
4. **O-NS:** a theorem stack connecting finite flow predicates and local capacity control to the existence/regularity structure required for Navier–Stokes.

Without these bridge theorems, no continuum, Yang–Mills, or Navier–Stokes closure claim is licensed by the present paper.

Theorem/Status Ledger

Label	Claim	Status
A1	Finite relational kernel, admissible tests, behavioral congruence, observational quotient	U
A2	Behavioral congruence is an equivalence relation	U
A3	Compression yields singular projection interfaces whenever token-distinct configurations are behaviorally indistinguishable	U
A4	Finite refinement stabilizes	U
A5	Survivor chains, one-step defect, and defect ledger are well-defined	U
B1	LS-2 locality-stability package	C
B2	Boundary determinacy from stabilized collar data under LS-2	C
C1	Outcome-count bound by collar-type count	C
C2	Boundary-capacity counting inequality and interface-capacity inequality	C
D1	Phase-A packing contradiction and local exclusion theorem	C
E1	Certificate instantiation lemma	U
E2	Regime-specific certification of named finite predicates	Comp
F1	Continuum realization theorem	Open
F2	Yang–Mills closure	Open
F3	Navier–Stokes closure	Open

Conclusion

We have developed a finite relational kernel in which observable structure is defined by admissible internal tests rather than primitive identity. Within that kernel, refinement stabilizes after finitely many steps, singular compression interfaces can be handled through class-level continuation and defect bookkeeping, and—under an explicit LS-2 locality package—stabilized local observables are determined by bounded collar data. This yields an interface-capacity inequality, and under an additional Phase-A multiplicity demand, a packing contradiction excluding capacity-exceeding local regimes.

The paper also isolates a disciplined computational layer: certificates verify named predicates in named finite regimes and support only regime-specific conclusions. Finally, it states the exact frontier. The finite theorem package is complete on its own terms, but every larger interpretation remains governed by explicit open bridge obligations.

A Appendix roadmap

The flagship paper is intentionally narrow. Technical expansions that do not alter the theorem spine belong in appendices or companion notes. In particular, the following materials are natural appendix targets:

1. extended proof expansions for refinement and ledger identities;
2. explicit collar-typing conventions and encoding choices;
3. certificate catalogs, hashes, and bounded-regime audit summaries;
4. optional toy examples illustrating singular projection interfaces or stabilization.

Material on sector algebra, screened sectors, Yang–Mills-type programs, Navier–Stokes-type programs, and broader interpretation belongs in companion papers rather than in the flagship theorem path.

B Appendix A: Proof expansion for finite stabilization

This appendix records the slightly more explicit proof logic behind the stabilization theorem.

Lemma B.1. *Let $P_0, P_1, P_2, \dots$ be a sequence of partitions of a finite set S such that each P_{n+1} refines P_n. Then there exists N such that $P_n = P_N$ for all $n \geq N$.*

Proof. Let $|S| = m$. Every partition of S has between 1 and m blocks, and in fact the total number of distinct partitions is finite (equal to the mth Bell number, though that exact value is not needed here). Because P_{n+1} refines P_n, the sequence cannot revisit a partition after a strict refinement has occurred unless it has already stabilized. Therefore an infinite strictly refining chain would produce infinitely many distinct partitions of S, impossible because only finitely many partitions exist. Hence the chain must eventually stabilize. □

Remark B.2. The main text uses only the existence of stabilization, not any explicit upper bound on the stabilization index. In bounded certified regimes, one may of course record an actual stabilization step computationally.

C Appendix B: Proof expansion for the boundary-capacity theorem

We expand the logical structure of the counting theorem in the order used by the flagship paper.

Lemma C.1. *Assume LS-2. If two admissible instances induce the same stabilized collar type on $\partial_{\rho^*}U$, then every realized stabilized local outcome determined on U is the same in both instances.*

Proof. This is exactly the boundary determinacy theorem from the main text, restated to emphasize that it identifies local outcome classes through collar data. □

Lemma C.2. *Assume LS-2. Then there is an injection*

$$\iota_U : \mathcal{O}(U) \hookrightarrow \mathcal{C}_{\rho^*}(U)$$

from realized stabilized local outcome classes into realized stabilized collar types.

Proof. Choose, for each realized stabilized local outcome class $o \in \mathcal{O}(U)$, one realized stabilized collar type $c \in \mathcal{C}_{\rho^*}(U)$ that gives rise to o. This defines a map on realized outcomes. If two distinct outcomes mapped to the same collar type, boundary determinacy would force them to coincide. Therefore the map is injective. □

Lemma C.3. *Let Σ_∂ be a collar alphabet of size K_0. Then for every finite region U,*

$$|\mathcal{C}_{\rho^*}(U)| \leq K_0^{|\partial_{\rho^*}U|}.$$

Proof. A raw symbolic boundary assignment chooses one symbol of Σ_∂ at each site of $\partial_{\rho^*}U$, so the total number of such assignments is $K_0^{|\partial_{\rho^*}U|}$. Realized stabilized collar types cannot outnumber raw assignments because each realized type is represented by at least one such assignment. □

Theorem C.4. *Assume LS-2 and a stabilized collar alphabet of size K_0. Then*

$$|\mathcal{O}(U)| \leq K_0^{|\partial_{\rho^*}U|}.$$

Proof. By the injective map above,

$$|\mathcal{O}(U)| \leq |\mathcal{C}_{\rho^*}(U)|.$$

By the collar-alphabet bound,

$$|\mathcal{C}_{\rho^*}(U)| \leq K_0^{|\partial_{\rho^*}U|}.$$

Combining the inequalities yields the claim. □

Corollary C.5. *Under the same hypotheses,*

$$\log |\mathcal{O}(U)| \leq |\partial_{\rho^*}U| \log K_0.$$

Proof. Take logarithms of the previous inequality. □

D Appendix C: Proof expansion for the packing contradiction

The contradiction theorem in the main text is intentionally short. For completeness we spell out the contradiction pattern one more time.

Expanded proof of the Phase-A packing contradiction. Assume LS-2, a stabilized collar alphabet of size K_0, and a Phase-A multiplicity demand $M(U)$ satisfying

$$M(U) > K_0^{|\partial_{\rho^*} U|}.$$

Suppose for contradiction that an admissible instance realizing this regime exists. Because the regime is assumed admissible and satisfies the Phase-A multiplicity demand, it must realize at least $M(U)$ distinct stabilized local outcome classes on U. Hence

$$|\mathcal{O}(U)| \geq M(U).$$

By the boundary-capacity theorem, however, every admissible instance under LS-2 satisfies

$$|\mathcal{O}(U)| \leq K_0^{|\partial_{\rho^*} U|}.$$

Combining the inequalities gives

$$M(U) \leq |\mathcal{O}(U)| \leq K_0^{|\partial_{\rho^*} U|},$$

contradicting the strict assumption

$$M(U) > K_0^{|\partial_{\rho^*} U|}.$$

Therefore no such admissible instance exists. □